

5.)

1) convert degree $\rightarrow$ rad

$$\begin{aligned}
 & -49^\circ \cdot \frac{\pi}{180^\circ} \\
 & = \frac{-49\pi}{180} \\
 & \approx -0.855 \text{ rad}
 \end{aligned}$$

2) convert degree $\rightarrow$ rad

$$\begin{aligned}
 & 3,400^\circ \cdot \frac{\pi}{180^\circ} \\
 & = \frac{3400\pi}{180} \\
 & \approx 59.341 \text{ rad}
 \end{aligned}$$

3) convert rad $\rightarrow$ degree

$$\begin{aligned}
 & -\frac{3\pi}{2} \cdot \frac{180^\circ}{\pi} \\
 & = -270^\circ
 \end{aligned}$$

4) convert rad $\rightarrow$ degree

$$\begin{aligned}
 & -1.9 \cdot \frac{180^\circ}{\pi} \\
 & = \frac{(-1.9)(180)}{\pi} \\
 & \approx -108.9^\circ
 \end{aligned}$$

5) convert rad $\rightarrow$ degree

$$\begin{aligned}
 & 3.4 \cdot \frac{180^\circ}{\pi} \\
 & = \frac{(3.4)(180)}{\pi} \\
 & \approx 194.8^\circ
 \end{aligned}$$

6) convert rad $\rightarrow$ degree

$$\begin{aligned}
 & \frac{19\pi}{10} \cdot \frac{180^\circ}{\pi} \\
 & = 342^\circ
 \end{aligned}$$

7) measure of angle in standard position is given. find 2 pos and neg angles that are coterminal with given angle.

$$\begin{array}{lcl}
 \frac{\pi}{4} & \frac{\pi}{4} + 2\pi = \frac{9\pi}{4} \text{ rad} & \frac{\pi}{4} + 4\pi = \frac{17\pi}{4} \text{ rad} \\
 & \frac{\pi}{4} - 2\pi = -\frac{7\pi}{4} \text{ rad} & \frac{\pi}{4} - 4\pi = -\frac{15\pi}{4} \text{ rad}
 \end{array}$$

8) measure of angle in standard position is given. find 2 pos and neg angles that are coterminal with given angle.

$$\begin{array}{lcl}
 -135^\circ & -135^\circ + 360^\circ = 225^\circ & -135^\circ + 720^\circ = 585^\circ \\
 & -135^\circ - 360^\circ = -495^\circ & -135^\circ - 720^\circ = -855^\circ
 \end{array}$$

9) measure of 2 angles in standard position given. Determine whether they are coterminal.

$$\begin{aligned}
 & -20^\circ, 340^\circ \\
 & 340^\circ - (-20^\circ) = 360^\circ \quad \leftarrow 360^\circ \text{ which is a full rotation} \\
 & \text{The angles are coterminal}
 \end{aligned}$$

10) measure of 2 angles in standard position given. Determine whether they are coterminal.

$$\begin{aligned}
 & \frac{4\pi}{5}, \frac{14\pi}{5} \\
 & \frac{14\pi}{5} - \frac{4\pi}{5} = \frac{10\pi}{5} = 2\pi \quad \leftarrow 2\pi \text{ which is a full rotation} \\
 & \text{The angles are coterminal}
 \end{aligned}$$

11) find angle between 0° and 360° that is coterminal with 371°

$$\begin{aligned}
 & 371^\circ - x = 360^\circ \\
 & x = 11^\circ
 \end{aligned}$$

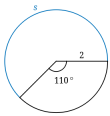
12) find angle between 0 and 2π that is coterminal with 27π

$$\frac{27\pi}{2\pi} = 13.5$$

$$13 \cdot 2\pi = 26\pi$$

$$27\pi - 26\pi = \boxed{\pi}$$

- 13) find length s of circular arc shown



$$s = r\theta$$

$$s = 2 \cdot ((360^\circ - 110^\circ) \cdot \frac{\pi}{180^\circ})$$

$$= 2(250 \cdot \frac{\pi}{180^\circ})$$

$$= 2(\frac{25\pi}{18})$$

$$s = \boxed{\frac{50\pi}{18}}$$

- 14) find radius r of circle with arc length s



$$s = r\theta$$

$$3 = r \cdot 1$$

$$r = \boxed{3}$$

- 15) central angle θ in circle of radius 3m is subtended by an arc of length 4m. find measure of θ in radians

$$s = r\theta$$

$$4 = 3\theta$$

$$\theta = \boxed{\frac{4}{3} \text{ rad}}$$

find measure of θ in degrees

$$\theta = \frac{4}{3} \cdot \frac{180^\circ}{\pi}$$

$$= \frac{240^\circ}{\pi}$$

$$\theta \approx \boxed{76.4^\circ}$$

- 16) Find area A of the sector

a)



$$60^\circ \cdot \frac{\pi}{360^\circ} = \frac{\pi}{3}$$

$$A = \frac{1}{2} r^2 \theta$$

$$= \frac{1}{2} 7^2 \left(\frac{\pi}{3} \right)$$

$$= \frac{1}{2} \cdot 49 \cdot \frac{\pi}{3}$$

$$= \frac{49\pi}{6}$$

$$A = \boxed{\frac{49\pi}{6} \text{ u}^2}$$

b)



$$A = \frac{1}{2} r^2 \theta$$

$$= \frac{1}{2} 15^2 (0.4)$$

$$A = \boxed{45 \text{ u}^2}$$

- 17) find area of sector with central angle $\frac{5\pi}{9}$ rad in circle of radius 20m

$$A = \frac{1}{2} r^2 \theta$$

$$= \frac{1}{2} 20^2 \left(\frac{5\pi}{9} \right)$$

$$\approx \boxed{349.1 \text{ m}^2}$$

- 18) area of sector of circle with central angle 130° is 50 m^2 . find radius of circle.

$$A = \frac{1}{2} r^2 \theta$$

$$50 = \frac{1}{2} r^2 \left(130^\circ \cdot \frac{\pi}{180^\circ} \right)$$

$$50 = \frac{1}{2} r^2 \left(\frac{13\pi}{18} \right)$$

$$r^2 = \frac{1800}{13\pi}$$

$$r \approx \boxed{6.6 \text{ m}}$$

19) car's wheels are 20 in. in diameter. how far in miles will car travel if it's wheels revolve 10,000 times without slipping?

$$d = 20 \text{ in}, r = 10 \text{ in}$$

$$10,000 \cdot 20\pi \text{ in} \cdot \frac{1 \text{ ft}}{12 \text{ in}} \cdot \frac{1 \text{ mi}}{5280 \text{ ft}} \approx 9.90 \text{ mi}$$

20) Memphis, Tennessee, and Merida, Yucatan, lie approx. on the same meridian. Memphis has a latitude of 35°N , and Merida has a latitude of 21°N . Find distance in mi. between the two cities. radius of Earth is 3960 mi.

$$s = r\theta$$

$$s = 3960 \cdot 14^\circ \cdot \frac{\pi}{180^\circ}$$

$$s \approx 968 \text{ mi}$$

21) wheels of a car have radius 12 in. and are rotating at 650 rpm. find speed of the car in mi/h.

$$\text{circumference of wheel: } 2\pi(12 \text{ in}) = 24\pi \text{ in}$$

$$\text{distance/minute: } 24\pi \frac{\text{in}}{\text{rev}} \cdot 650 \frac{\text{rev}}{\text{min}} = 15600\pi \frac{\text{in}}{\text{min}}$$

$$\text{Speed: } \frac{15600\pi \frac{\text{in}}{\text{min}}}{325} \cdot \frac{60 \text{ min}}{1 \text{ hr}} \cdot \frac{1 \text{ ft}}{12 \text{ in}} \cdot \frac{1 \text{ mi}}{5280 \text{ ft}} = \frac{325\pi}{22} \text{ mi/h}$$